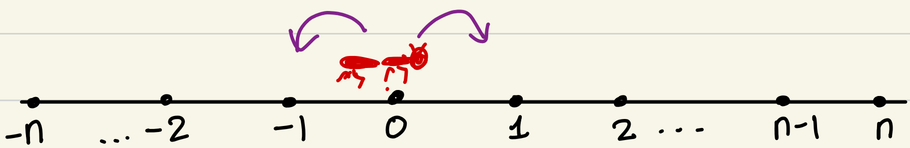
21 JULY
LECT 1.

PROBABILITY 3

$\boxed{1}$ Let us start with an example you might have seen in your previous probability courses:

n - step random walk on $\mathbb{Z}$



- A drunk ant is performing a random walk in 1 dim.

- Starting at 0, at each time step (up til n) it is equally likely to take one step forward or one step backward.

- Q: What is the probability that after n steps it finds himself at 0, which is where it started?

$\boxed{2}$ Before answering this qt., we observe that this random experiment falls in the framework of :

<u>Discrete probability space</u>. is a pair (Ω, q) where

- Ω = countable set called the sample space (this contains all possible outcomes of a random experiment).

- $q : \Omega \longrightarrow [0,1]$ is a function s.t.

$$\sum_{\omega \in \Omega} q(\omega) = 1.$$

Model or Axioms

- Any subset $A \subseteq \Omega$ is called an event & the probability of A,

$$\mathbb{P}(A) := \sum_{\omega \in A} q(\omega).$$

$\boxed{3}$ Coming back to the qt. in $\boxed{1}$

$$\Omega = \{ x = (x_0, x_1, x_2, \ldots, x_n) : x_0 = 0 \, , \, x_{i+1} - x_i \in \{\pm 1\} \; \forall \, 0 \le i \le n-1 \}$$

Assuming independence of the steps (which we haven't defined so far), a reasonable q is

$$q(x) = \frac{1}{2^n}.$$

$$A = \{x \in \Omega : x_n = 0\}.$$

$$\mathbb{P}(A) = \sum_{x \in A} q(x) = \frac{|A|}{2^n} = \frac{\binom{n}{n/2}}{2^n}.$$

$\boxed{4}$ Suppose in the example $\boxed{1}$, the ant continues the random walk indefinitely.

Let us see if this falls in the framework of a discrete prob. space...

$$\Omega = \{ \underline{x} = (0, x_1, x_2, \dots) : x_{i+1} - x_i \in \{+1, -1\} \; \forall i \geq 0 \}.$$

<u>Three Qts.</u>

(i) What would you want to assign $q(x)$?
$\| \; q(x) = 0$ is reasonable $\|$

(ii) Cardinality of Ω?

- Uncountable
- There is a 1-1 correspondence b/w

$$\Omega \longleftrightarrow S_1 = \{ y = (y_1, y_2, \ldots) : y_i \in \{+1, -1\} \}.$$

$$(x_1, x_2, \ldots) \longleftrightarrow (x_1, x_2 - x_1, x_3 - x_2, \ldots).$$

- $S_2 = \{ z = (z_1, z_2, \ldots) : z_i \in \{0, 1\} \}$

$$\Big\updownarrow \text{ 1-1 correspondence}$$

$2^{\mathbb{N}}$ = Subsets of $\mathbb{N}$.

- We know that $2^{\mathbb{N}}$ is not countable.

[For any set S, there is no bijection b/w S and 2^S for the following reason.

Suppose $\exists$ a bijection $\varphi : S \longrightarrow 2^S$.

Consider $A \subseteq S$ defined by:

$$A = \{ x \in S : x \notin \varphi(x) \}.$$

Let $a \in S$ be s.t. $\varphi(a) = A$.

Two possibilities :

① $a \in A$ or $a \notin A$

- Can't happen by defn. of A

- If this happens, then $a \in A$ $(\Rightarrow\Leftarrow)$

Both possibilities lead to a contradiction.]

iii) What is the probability that at the 5^{th} step & 7^{th} step the ant took a step fwd?

Call this event E.

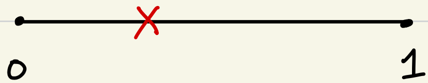
One would want to answer $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.

But if this were to fit into the framework of a discrete probability space,

$$\frac{1}{4} = \mathbb{P}(E) \overset{?}{=} \underbrace{\sum_{\omega \in E}}_{\text{uncountable sum?}} \underbrace{q(\omega)}_{=0}$$

So on several counts, the random experiment fails to fall in the framework of a discrete prob. space.

5 Another example



Break a stick of length 1 at a random point.

- If we ask: What is the prob. that the point B. is bet. 0.6 and 0.7?

We would want to say 0.1.

- $\mathbb{P}(B=0.5) = 0.$
- $\mathbb{P}(B \text{ is a rational/irrational point}) = ?$

$\mathbb{P}(B \in \mathcal{C}) = ?$

Cantor set

Here $\Omega = \{x : x \in [0,1]\}$ ~ Again an uncount. set.

Let us take a detour & look at the DPS:

Equivalent to a DPS is a "probability space" which is a tuple $(\Omega, \mathcal{F}, \mathbb{P})$ where

Ω = sample space (which in case of DPS is a countable set)

$\mathcal{F}$ = Events (all subsets of Ω).

$\mathbb{P} : \mathcal{F} \longrightarrow [0,1]$ satisfying:

① $\mathbb{P}(\Omega) = 1$.

② (Countable additivity) If A_n's are subsets of Ω which are disjoint, then

$$\mathbb{P}(\cup A_n) = \sum_n \mathbb{P}(A_n).$$

Let's take this point of view & ask if we can define

$$\mathbb{P} : 2^{[0,1]} \longrightarrow [0,1] \quad \text{s.t.}$$

(Power set of $[0,1]$, i.e., set of all subsets of $[0,1]$)

- $\mathbb{P}([a,b]) = b-a$, &

- $\mathbb{P}$ satisfies countable additivity...

Q: Is there a nice candidate for $\mathbb{P}$? Here is one... called the outer measure:

For any $S \subseteq [0,1]$, and a countable collection of intervals $\{I_k\}_k$, each $I_k \subset [0,1]$, we say that I_k's cover S if

$$S \subseteq \bigcup_{k=1}^{\infty} I_k.$$

$$\mathbb{P}_*(S) := \inf \left\{ \sum_{k=1}^{\infty} |I_k| : \{I_k\} \text{ is a cover for } S \right\}.$$

- What is $\mathbb{P}_*(\mathbb{Q} \cap [0,1]) = 0$.

- What is $\mathbb{P}_*([a,b]) = b-a$.

We will show later that $\exists A \subseteq [0,1]$ s.t. $\mathbb{P}_*(A) = 1$ and $\mathbb{P}_*(A^c) = 1$ & hence it violates countable (indeed finite) additivity.

So although $\mathbb{P}_*$ looks like a good candidate, it is not!

But are there other $\mathbb{P}$'s which satisfy the reqd. condt. ?

To be proved soon

- There is no $\mathbb{P}$ defined on all of $2^{[0,1]}$ satisfying countable additivity and $\mathbb{P}([a,b]) = b-a$.

- Although there is a $\mathbb{P}$ on $2^{[0,1]}$ which is finitely additive ...

$\boxed{6}$ Solution for this predicament : Is to restrict the events to a smaller subset of $2^{[0,1]}$ and ask for $\mathbb{P}$ satisfying

- countable additivity & • $\mathbb{P}([a,b]) = b-a$.

Def. A probability space is a tuple $(\Omega, \mathcal{F}, \mathbb{P})$

- Ω is a non-empty set called the sample space.

- $\mathcal{F}$ is a σ-algebra on Ω, i.e., $\mathcal{F} \subseteq 2^{\Omega}$ (i.e., it is a collection of subsets of Ω) s.t.

  i) $\phi, \Omega \in \Omega$

  ii) $A \in \Omega \Rightarrow A^c \in \Omega$

  iii) $A_n \in \Omega \Rightarrow \cup A_n \in \Omega$.

$\mathcal{F}$ consists of events/measurable sets and these are the only sets we are interested in...

- $\mathbb{P} : \mathcal{F} \to [0,1]$ called probability measure is s.t.:

  i) $\mathbb{P}(\Omega) = 1$

  ii) Countably additivity: If $A_n \in \mathcal{F}$ and are mutually disjoint, then

$$\mathbb{P}(\cup A_n) = \sum \mathbb{P}(A_n).$$

Def [Measure space]

$\boxed{7}$

Examples of σ-algebras $\mathcal{F}$. (on Ω which is an uncountable set).

- $\mathcal{F}_1$ = power set 2^{Ω}.

- $\mathcal{F}_2 = \{A \subseteq \Omega : A \text{ is finite or } A^c \text{ is finite}\}$

NOT a σ-algebra.

- Modification of $\mathcal{F}_2$:

$$\mathcal{F}_3 = \{A \subseteq \Omega : A \text{ is countable or } A^c \text{ is countable}\}.$$

$\boxed{8}$ Claim: Let Ω be a non empty set; and $S \subseteq 2^{\Omega}$. Then we can make sense of the smallest sigma algebra containing S, also denoted the sigma algebra generated by S $\sigma(S)$.

Lemma: Let $\Omega \neq \phi$. Consider

$$\sigma(S) := \bigcap_{\alpha} \{\mathcal{F}_\alpha : \mathcal{F}_\alpha \text{ is a } \sigma\text{-algebra on } \Omega \text{ \& } \mathcal{F}_\alpha \supseteq S\}.$$

Then $\sigma(S)$ is a σ-algebra and contains S.

<u>Pf. of lemma</u>:

- That $S \subseteq \sigma(S)$ is clear.
- $\Omega \in$ each $\mathcal{F}_\alpha \Rightarrow \Omega \in \sigma(S)$.
- If $A \in \sigma(S) \Rightarrow A \in$ each $\mathcal{F}_\alpha \Rightarrow A^c \in \mathcal{F}_\alpha, \forall \alpha$

$$\Rightarrow A^c \in \cap \mathcal{F}_\alpha.$$
$$\Rightarrow A^c \in \sigma(S).$$

- If $\{A_n\}_{n=1}^{\infty}$ is s.t. each $A_n \in \sigma(S)$

$$\Rightarrow \forall \alpha: \{A_n \in \mathcal{F}_\alpha \ \forall n\} \Rightarrow \forall \alpha: \ \cup A_n \in \mathcal{F}_\alpha$$

$$\Rightarrow \cup A_n \in \cap \mathcal{F}_\alpha \Rightarrow \cup A_n \in \sigma(S).$$
